

# CSE 3112 Final

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## I. QUESTION (20 POINTS)

Give a big-Oh characterization, in terms of  $N$ , of the running time of the following functions. Show your work.

```
a.)
int magic1(int N){
    int i, count = 1;
    while (N > 1){
        count++;
        N = N / 2;
    }
    return count;
}

b.)
int magic2(int N){
    if (N == 1)
        return 1;
    else
        return 1 + magic2(N / 2);
}
```

## II. QUESTION (15 POINTS)

A word can be changed to another word by a 1-character substitution. Assume that a dictionary of 5-letter words exists. Give an algorithm to determine if a word A can be transformed to a word B by a series of 1-character substitutions, and if so, outputs the corresponding sequence of words. Note that all intermediate words must be in the dictionary. As an example, bleed converts to blood by the sequence bleed, blend, blond, blood. Write a greedy algorithm for this problem.

## III. QUESTION (15 POINTS)

Write a divide and conquer algorithm for finding all the elements in a sorted array whose values fall between two given values L and U (inclusively)?

## IV. QUESTION (15 POINTS)

Modify Dijkstra's shortest path algorithm with nonnegative numbers assigned to its vertices so that

the length of a path defined as the sum of the vertex numbers on the path. Use adjacency matrix representation of the graph.

## V. QUESTION (20 POINTS)

Given the adjacency matrix representation of a graph  $G$ , write a method to return the adjacency list representation of  $G$ .

## VI. QUESTION (15 POINTS)

A rook stands on the upper left square of an  $M \times N$  rectangular chess board. Two players make turns moving the rook either horizontally to the right or vertically downward. The player who can place the rook on the lower right square of the chessboard wins. Write a dynamic programming algorithm to find the winner depending on the size of the chess board.